HX-200 Bench Calibrator: Maintenance Manual

This manual covers routine maintenance of the HX-200 bench calibrator built by Halden Instruments. It is written for service technicians who already hold a level 2 electrical safety certificate. Read the safety section before opening the enclosure. Revision R4 replaces R3 from 2024-03-18.

1. Safety

The calibrator stores energy in its output capacitors for up to ninety seconds after the mains switch is turned off. Always wait two minutes, then confirm with a meter that the voltage across terminals T1 and T2 is below 5 V before touching any internal part. Never bypass the interlock on the rear panel: it is the only protection against the 1,000 V reference output.

- Wear insulated gloves rated for 1,000 V when the enclosure is open.
- Disconnect the instrument from the mains and from any device under test.
- Keep the service log within reach and record every intervention.
- Report any burnt smell, discoloured board or swollen capacitor to the service lead.

2. Scheduled maintenance

Preventive maintenance follows the operating hours shown on the status screen. Intervals assume a clean laboratory at 23 degrees and 45 percent relative humidity. Halve every interval when the instrument works in a production line or in a dusty workshop, because dust on the reference board shifts the readings long before it triggers an alarm.

Task	Interval	Part number	Time
Clean air filter	500 h	HX-FLT-021	10 min
Replace fan assembly	8,000 h	HX-FAN-118	35 min
Replace backup battery	24 months	BAT-CR2032-HX	15 min
Verify reference output	12 months	CAL-KIT-07	2 h
Replace output relay	100,000 cycles	RLY-4471-B	40 min

The time column is the hands-on time for a trained technician. It does not include the warm-up period of forty-five minutes that the instrument needs before any verification step.

2.1 Cleaning the air filter

Remove the four screws of the rear grille with a T10 driver. Slide the filter HX-FLT-021 out of its rail and blow it clean from the inside towards the outside with dry compressed air at no more than 2 bar. Replace it when the foam is torn or when it stays grey after cleaning. Refit the grille and tighten the screws to 0.6 N·m. Record the cleaning in the service log with the operating hours.

2.2 Replacing the fan assembly

A worn fan announces itself with a rising pitch and with the warning W-212 on the status screen. Order the fan assembly HX-FAN-118, which includes the guard and the connector. Unplug the fan cable from connector J14 on the power board, remove the two retaining clips and lift the old fan out. The new fan must blow towards the rear of the case; an arrow on its frame shows the airflow.

3. Verification of the reference output

Verification compares the output of the calibrator with a reference standard whose uncertainty is at least four times smaller. Use the calibration kit CAL-KIT-07 and a reference multimeter that has been calibrated within the last twelve months. Verify the seven points of the table below, in the listed order, after the forty-five minute warm-up.

Point	Nominal	Tolerance	Range
P1	10 mV	±2 µV	100 mV
P2	100 mV	±8 µV	100 mV
P3	1 V	±25 µV	1 V
P4	10 V	±180 µV	10 V
P5	100 V	±2.1 mV	100 V
P6	500 V	±12 mV	1,000 V
P7	1,000 V	±26 mV	1,000 V

If a point falls outside its tolerance, repeat it once after five minutes. If it fails again, the instrument must be adjusted by an accredited laboratory; field adjustment voids the certificate issued with the instrument. Attach the verification record to the service log and label the instrument with the date of the next verification.

4. Troubleshooting

Most faults reported by customers come from three sources: a blocked air filter, a dead backup battery or a worn output relay. The status screen shows a code for each of them. Start with the code, follow the table in this section and only then open the enclosure.

- W-212 fan speed low: clean the filter, then replace HX-FAN-118 if the warning stays.
- E-301 clock lost after power off: replace the backup battery BAT-CR2032-HX.
- E-415 output relay contact resistance high: replace relay RLY-4471-B.
- E-520 reference drift: verify the output; if P3 or P4 fail, send the unit for adjustment.

When none of these codes is shown and the readings are still unstable, check the grounding of the bench first. A floating ground can add noise of several microvolts on the lower ranges, which looks like a fault in the instrument but disappears as soon as the bench is bonded to earth.

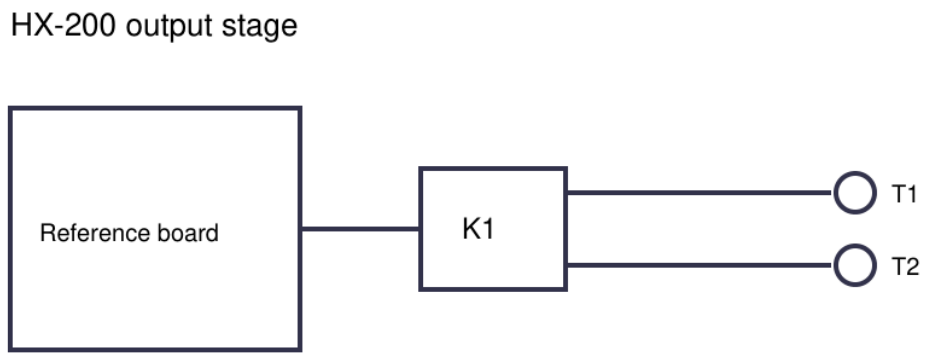

Figure 1. Simplified wiring of the output stage, with relay K1 between the reference board and terminals T1 and T2.

5. Spare parts and support

Spare parts are shipped from the central warehouse within two working days. Quote the serial number of the instrument, which starts with HX2 followed by seven digits, when you order. Support tickets are answered by the service desk from Monday to Friday. Urgent tickets about a safety issue are answered within four hours.